the spectrum, adjacent to the red, expecting to read no more than room temperature throughout the experiment. But that's not what happened. The temperature of his control thermometer rose even higher than in the red.

Herschel wrote:

> [I] conclude, that the full red falls still short of the maximum of heat; which perhaps lies even a little beyond visible refraction. In this case, radiant heat will at least partly, if not chiefly, consist, if I may be permitted the expression, of invisible light; that is to say, of rays coming from the sun, that have such a momentum as to be unfit for vision.[††]

Holy s#%t!

Herschel inadvertently discovered "infra" red light, a brand-new part of the spectrum found just "below" red, reported in the first of his four papers on the subject.

Herschel's revelation was the astronomical equivalent of Antonie van Leeuwenhoek's discovery of "many very little living animalcules, very prettily a-moving"[†††] in the smallest drop of lake water. Leeuwenhoek discovered single-celled organisms—a biological universe. Herschel discovered a new band of light. Both hiding in plain sight.

Other investigators immediately took up where Herschel left off. In 1801 the German physicist and pharmacist Johann Wilhelm Ritter found yet another band of invisible light. But instead of a thermometer, Ritter placed a little pile of light-sensitive silver chloride in each visible color as well as in the dark area next to the violet end of the spectrum. Sure enough, the pile in the unlit patch darkened more than the pile in the violet patch. What's beyond violet? "Ultra" violet, better known today as UV.

Filling out the entire electromagnetic spectrum, in order of low-energy and low-frequency to high-energy and high-frequency, we have: radio waves, microwaves, infrared, ROYGBIV, ultraviolet, X-rays, and gamma rays. Modern civilization has deftly exploited each of these bands for countless household and industrial applications, making them familiar to us all.

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After the discovery of UV and IR, sky-watching didn't change overnight. The first telescope designed to detect invisible parts of the electromagnetic spectrum wouldn't be built for 130 years. That's well after radio waves, X-rays, and gamma rays had been discovered, and well after the German physicist Heinrich Hertz had shown that the only